

First Perturb-seq Experiments in Two Yeasts

A bench plan for *S. cerevisiae* and *P. kudriavzevii*

Michael Volk

September 15, 2026

Abstract

Two first runs are ready to start, one per host, and they do not depend on each other. In *S. cerevisiae* the guide library already exists, so the open question is whether perturbation identity can be read out of a cell at all, and one droplet channel answers it for \$4,493. In *P. kudriavzevii* no pooled library is reachable yet, so the first run carries no guides and instead measures whether the assay works on that cell wall, which is worth doing across several growth media rather than one. Both are sized against a published genome-scale yeast screen whose data, on reanalysis, does not reproduce itself, and the three checks that would have caught that are built into the designs here.

How to read this. Section 1 is the plan and the costs. Section 2 is what the bench actually does on each of the two routes. Section 3 is the published screen and what went wrong with it, which is where the quality checks in section 5 come from. Section 4 answers the two decisions that were open: media against readout, and plasmid copy number.

Costing and method review: `notes-tex/024-perturb-seq-costing/`

Generating scripts: `experiments/024-perturb-seq-costing/scripts/pilot\_options.py`

References: Zotero `torchcell/microbe-perturb-seq` (group $\cup$ personal)

Section status: ✔ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

Provenance: ▲ not in the library mirror; verify at the linked source ● from a review's summary table, not the primary paper

Contents

1	The Plan ✗	1
2	The Two Bench Routes ✗	3
3	The Published Screen, and Why We Do Not Trust It ✗	4
4	The Two Decisions That Were Open ✗	7
4.1	Growth Media, or the Readout First ✗	7
4.2	Plasmid Copy Number ✗	7
5	Running It ✗	8
5.1	What Each Run Must Return ✗	8
5.2	Three Things to Get Right at the Bench ✗	8
5.3	What Happens Next, Under Each Outcome ✗	8

1 The Plan ✗

Two runs, one per host, started in parallel by two people. Neither waits on the other, and they answer different questions because the two organisms are at different stages.

***S. cerevisiae*: the library exists, so measure the readout.** A genome-wide single-guide CRISPR interference (CRISPRi) library is already in hand, six guides per gene over roughly 6,000 genes [1]. What has never been established is whether a cell's guide can be recovered from its RNA. Call that probability q , the fraction of cells whose perturbation is read. One channel on the core's 10x Genomics 5' kit answers it (section 2).

***P. kudriavzevii*: no library is reachable, so measure the assay.** Transformation efficiency in this host is three to five orders of magnitude below *S. cerevisiae*, which puts a pooled library-scale screen out of reach until that moves. A first run therefore carries no guides at all, and what it returns is whether the cell wall yields to in-droplet digestion, how many messenger RNA molecules are recovered per cell, and what fraction of reads is ribosomal. None of the three needs genetics.

Table 1: The candidate first runs, priced end to end. Usable cells are those expected to survive quality control, 55% of a droplet channel’s 20,000 and 25% of a protocol run’s 480,000. Recurring cost is reagents plus sequencing at the cheapest configuration the Carver Center sells that holds the reads, which at this scale is never the flow cell a genome-scale screen would use. The one-time column is the split-pool barcode plate set, which covers about 215 protocol runs. The route is the plate one for the eight-media row and 10x droplet for every other.

run	host	usable cells	recurring	one-time	returns
Guide capture, 1 channel	<i>S. cerevisiae</i>	11,000	\$4,493	–	q, the per-cell guide detection rate
Guide capture, low vs high copy	<i>S. cerevisiae</i>	22,000	\$8,576	–	q on two vector backbones
Unperturbed profile, 1 medium	<i>P. kudriavzevii</i>	11,000	\$3,810	–	wall digestion, UMIs per cell, rRNA fraction
Unperturbed profile, 4 media	<i>P. kudriavzevii</i>	44,000	\$12,550	–	the same three, plus a 4-condition response atlas
Plate route, 8 media in one run	<i>P. kudriavzevii</i>	20,000	\$4,990	\$10,000	the same three, across 8 media, plus the plate chemistry

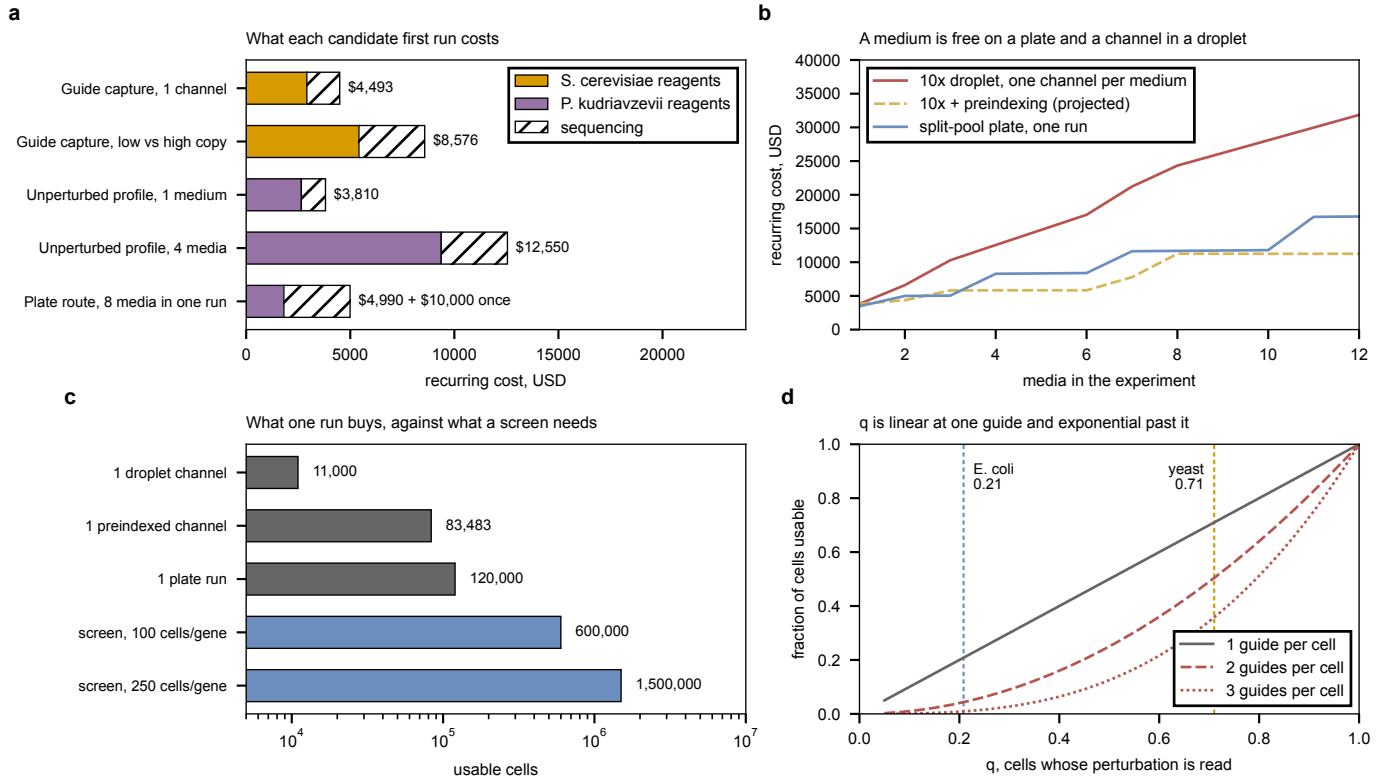


Figure 1: The two first runs cost under \$9,000 together, and the cheaper way to add a growth medium is a plate rather than a second droplet channel. (a) Recurring cost of each candidate, split into reagents (solid, by host) and sequencing (hatched). Sequencing is priced on the cheapest configuration the Carver Center sells that holds the reads, which at this scale is a MiSeq run or a split NovaSeq lane rather than the large flow cell a genome-scale screen fills. The plate route carries a \$10,000 one-time purchase of barcoded oligo plates, which covers about 215 protocol runs. (b) Recurring cost against the number of growth media, at one droplet channel’s worth of cells per medium. An unmodified droplet channel carries nothing that says which medium a cell came from, so each medium buys its own channel, at \$2,650 for the first and \$2,240 for each of the next six. Both plate routes read a well index out of every cell already, so media ride together in one run and the steps are sublibrary and run rounding rather than per-medium cost. (c) What one run buys against what a screen needs, on usable cells. (d) Why q is the quantity worth measuring first. At one guide per cell it scales the yield linearly; at k guides per cell the usable fraction is q^k , so the two published microbial values, 0.21 in *E. coli* [2] and more than 0.71 in *S. cerevisiae* [3], are the difference between multiplexing being expensive and being impossible. Generated by `experiments/024-perturb-seq-costing/scripts/pilot_options.py`.

What the prices do and do not include. Reagents and sequencing at on-campus rates, effective 2026-08-01. Not included: strain construction, the media themselves, staff time, and the cell counter discussed in section 5.2. The sequencing line is the one most likely to move, because a lane is an indivisible purchase and a run that half-fills one pays for the whole thing.

Start both. They cost \$8,303 together and answer different questions. In *S. cerevisiae*, one 5' channel on the existing CRISPRi library measures q , the fraction of cells whose guide is readable, at \$4,493. In *P. kudriavzevii*, one channel with no library measures wall digestion, messenger RNA per cell, and ribosomal fraction, at \$3,810. Neither depends on the other, one person can own each, and the two results together decide every design question after them.

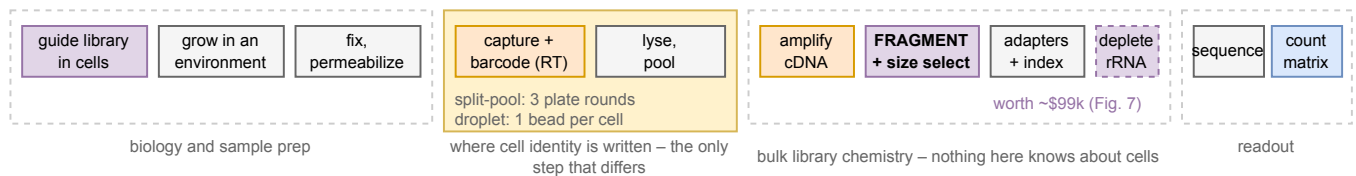
2 The Two Bench Routes x

Both routes end in the same object, a table of cells by genes with a perturbation label on each cell. They differ in how a cell keeps its identity while its RNA is being converted to a sequencing library, and that difference decides who does the work.

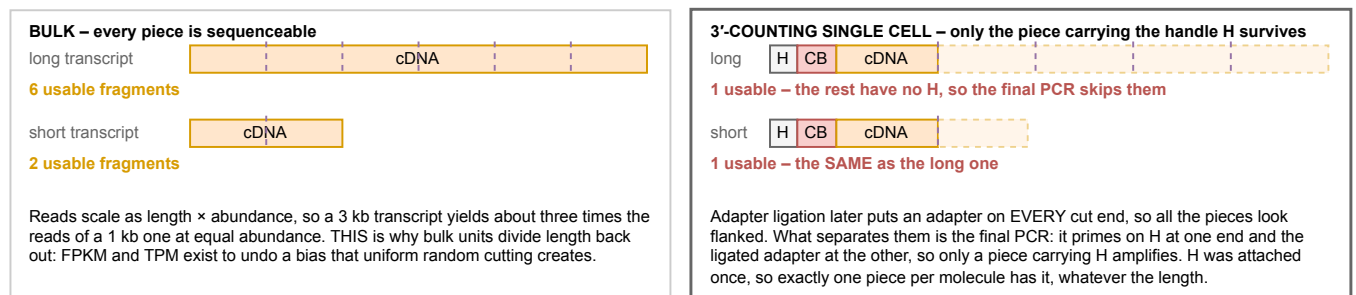
The whole experiment, and where each stage can lose you cells, molecules or identity

Read left to right. The wet-lab stages are the same for both method families; what differs is only how cells are kept apart while their barcodes are written (Figs. 2, 4 and 5). Fragmentation is drawn explicitly because it is the step that decides whether a count means "molecules" or "molecules times length", and because ribosomal RNA depletion is placed relative to it (Fig. 7).

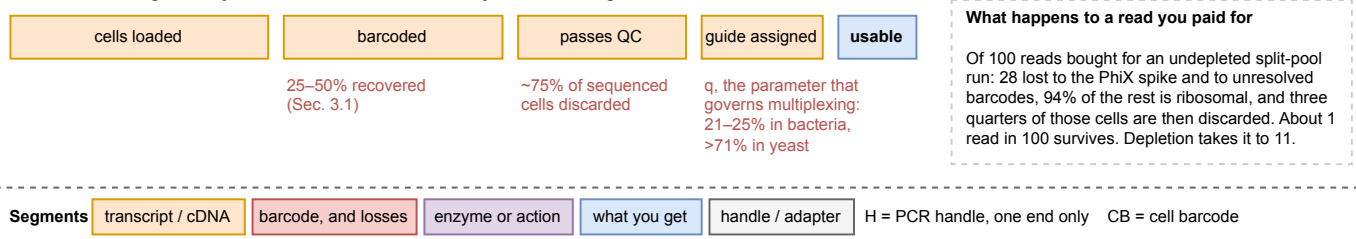
a The stages. Only the shaded block differs between split-pool and droplet.



b Fragmentation in bulk RNA-seq and in a 3'-counting single-cell assay. Same cutting, opposite consequence.



c What each stage costs you. The funnel, not the chemistry, is what a budget is made of.



Panel b is the reason this figure exists. Bulk and single-cell protocols perform the SAME physical cutting, and the consequence inverts because of what happens afterwards: in bulk every piece can be sequenced, so reads scale with length; in a 3'-counting assay only the piece carrying the outer handle can be, so exactly one survives per molecule regardless of length. Applying TPM or FPKM to a 10x or SPLiT-seq count matrix is therefore an error, not a stylistic choice.

Figure 2: The whole experiment, from library to count matrix, with the step each route replaces marked. The pooled library is transformed once and grown under the chosen condition; from there the droplet route isolates single cells in an emulsion and the plate route keeps them intact while barcoding them combinatorially across wells. Everything downstream of library preparation is shared, including the sequencer and the alignment. Redrawn from Fig. 5 of the costing review; source [notes/assets/drawio/perturb-seq-workflow.drawio.svg](#).

Droplet, which the core runs as a service. A microfluidic chip co-encapsulates one cell with one barcoded bead, and every molecule released inside that droplet picks up the bead's barcode. The Carver Center owns the instrument, the GEM-X 5' kit and the quality control, so our bench work ends when fixed cells are delivered. Yeast needs one modification, roughly a microlitre of zymolyase in the reaction to digest the cell wall [4], and the core has said it accepts that. A channel recovers about 20,000 cells and costs \$2,650 for the first library and \$2,240 for each of the next six.

Plate, which we would run ourselves. Cells are fixed and permeabilized, then distributed across a 96-well plate where each well appends a different barcode; the wells are pooled, re-split and barcoded again, twice more. A cell's identity is the sequence of wells it visited, so no instrument ever has to isolate it [5]. One pass through the protocol handles up to 96 samples and, at a full plate, roughly 480,000 cells. The reagents are cheap and the labor is not: about a week of skilled bench work per run, against a channel the core runs.

The plate's real advantage is the well index. Round one is a plate of wells, and its index is read out of every cell whether or not it has been given a meaning. Growing 96 conditions in a deep-well plate and barcoding them into the same run therefore costs no barcode space at all (fig. 1b). A droplet channel has no equivalent, so each condition buys its own channel. That single property is what makes the plate route the one to use when the experiment is about growth media, and it is why the *P. kudriavzevii* run should not be a single condition.

The guide readout, which is the whole *S. cerevisiae* question. Guide RNAs are transcribed by RNA polymerase III, so they are neither capped nor polyadenylated, and an assay that captures messenger RNA by its poly(A) tail sees nothing. That is why our library is read today by amplifying the plasmid rather than the RNA. The 5' kit may not have the problem: it captures the guide with a primer against the guide's own scaffold, carried in the reverse-transcription mix, and 10x state the requirement as an architectural one, that guides be "engineered for use with standard Cas9 systems with a protospacer on the 5' end"▲[external: 10x CG000735]. A conventional polymerase III cassette, including ours, has exactly that architecture.

Hypothesis (untested): the scaffold primer reverse transcribes a yeast polymerase III guide, and perturbation identity is therefore readable from an unmodified library. Two things have to hold and neither is established. The primer set is proprietary and validated against the standard *SpCas9* scaffold, and 10x say plainly that "alternate sgRNA structures ... have not been tested"▲[external: 10x CG000735], so our scaffold has to match theirs. And the poly-dT half of that primer mix contributes nothing to a guide with no tail, so capture rests entirely on the scaffold primer. If it fails, the fallback is a polyadenylated guide barcode transcribed by polymerase II, which is a construction project of several months.

The guide library is sequenced far more shallowly than the transcriptome, 5,000 read pairs per cell against 20,000, and 10x warn against sequencing it alone▲[external: 10x CG000735], so it is pooled with the expression library rather than submitted separately. It is in the price in table 1.

One channel, one existing library, and the answer is a number. The *S. cerevisiae* run is worth starting before anything else because it is the cheapest way to find out whether the whole program needs a new construct. If q comes back near the 0.71 a published yeast screen reports, single-guide screening works today and multiplexing is merely expensive. If it comes back near zero, the 5' scaffold primer does not engage a polymerase III transcript and a polyadenylated guide barcode has to be built before any screen is designed.

3 The Published Screen, and Why We Do Not Trust It x

The closest published experiment to the one proposed here is a genome-scale single-cell screen of the yeast knockout collection, about 3,100 deletions profiled by single-cell RNA sequencing [3]. It is the design this plan would otherwise copy. Loading it into our database and comparing it against the knockout expression data already there showed it does not agree with them, and then that it does not agree with itself.

This matters at the bench rather than in the analysis. Every failure below is one a run can be designed to catch, and all three checks are cheap.

First, what agreement looks like when it works. Two microarray studies of the same deletion collection, measured years apart in different hands [6, 7], agree on a deletion's expression profile at a median correlation of **0.74** over the 82 deletions they share, and on a reporter gene's behavior across deletions at 0.62. That is the reference number: a real knockout signature survives a change of laboratory.

Against that, the single-cell panel reads as noise. Over the 914 deletions it shares with the microarrays, the median agreement is **0.003**. Recomputing the fold change from the raw counts, rather than taking the released values, moves it to 0.013. The agreement does not improve with the number of cells behind a profile, which is what an undersampled but real measurement would do.

The panel does not reproduce itself, which is the finding that settles it. A cross-study zero cannot be interpreted without a within-study replicate, and the published work reports none. One is recoverable, because the same genotype's cells fall in several capture cartridges. Comparing a genotype against itself across two cartridges, each referenced to wild-type cells from its own cartridge, gives a median correlation of **0.043** over 17,554 pairs. Two *different*

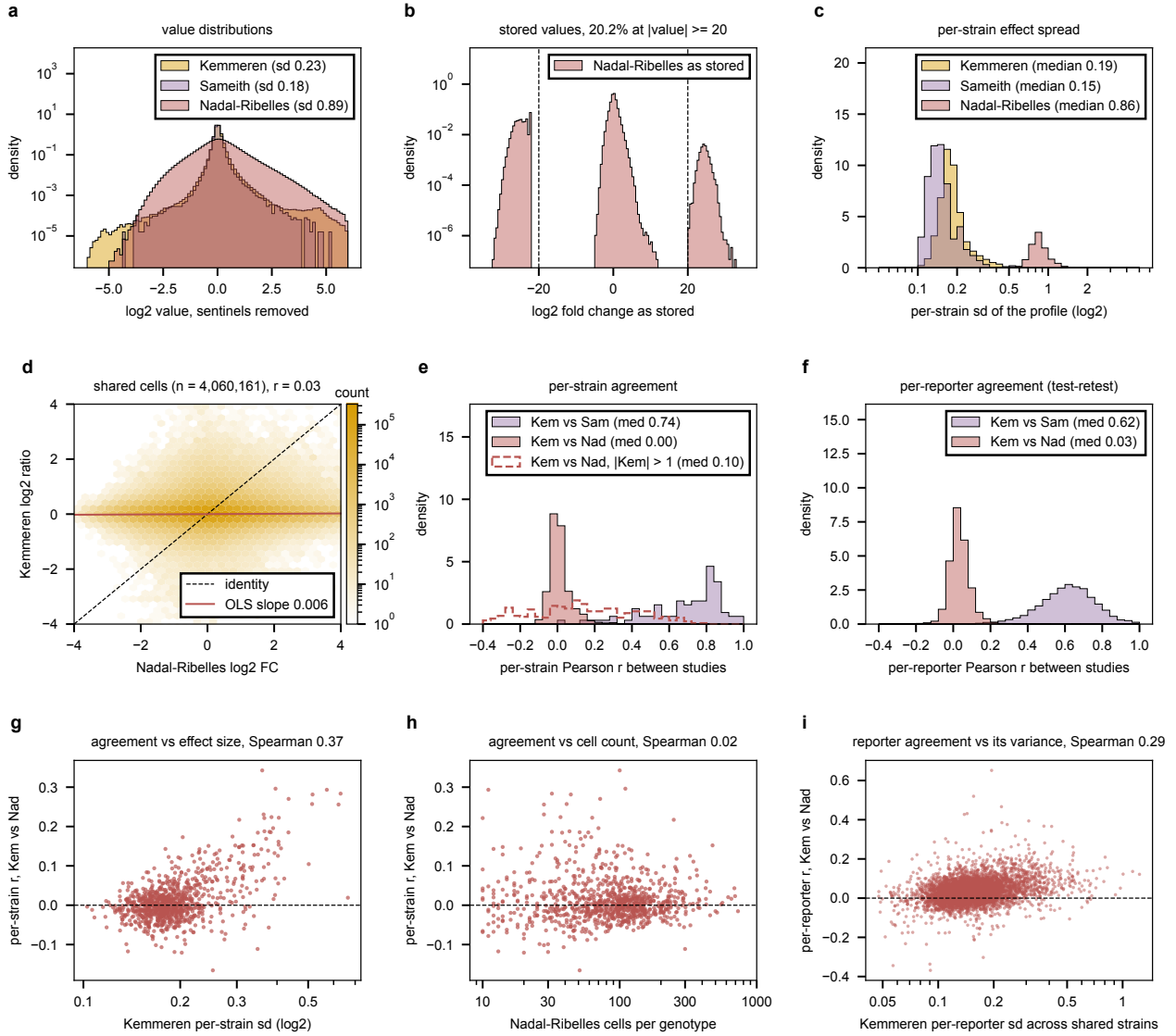


Figure 3: A deletion's microarray profile reproduces in a second microarray study at a median correlation of 0.74 and in the single-cell panel at 0.00. (a, b) The released single-cell values sit on a different scale, and one cell in five is a sentinel produced when a gene's mean is zero in one group, which the fold-change formula returns as roughly ± 23 . Those are discarded before every comparison here. (c) Spread of each deletion's profile. (d) Every shared deletion-by-gene value, with the fitted slope. (e) The headline: per-deletion agreement for the two microarrays against the microarray-to-single-cell comparison. (f) The same per reporter gene. (g, h, i) What the agreement tracks. It rises with how strongly a deletion responds and with how variable a reporter is, and it does not track cell count. Generated by `experiments/028-knockout-expression/scripts/cross_study\ko_expression.py`.

genotypes compared the same way give 0.022. Splitting one genotype's cells in half within a single cartridge gives 0.080 against a 0.051 null.

A genotype's profile reproduces itself about 0.02 above an unrelated genotype. Nothing downstream of that can agree with anything.

Two related traps are worth naming because both are easy to repeat. Referencing every profile to one pooled wild type instead of a per-batch wild type raises both the real and the null value to 0.11, which looks like signal and is a batch effect: wild type against wild type between cartridges differs by a standard deviation of 0.75 in $\log_2$, with 16% of genes beyond two-fold. And sharing the reference between two halves of a split-half test inflates it to 0.37, which is the reference's own sampling noise measured twice.

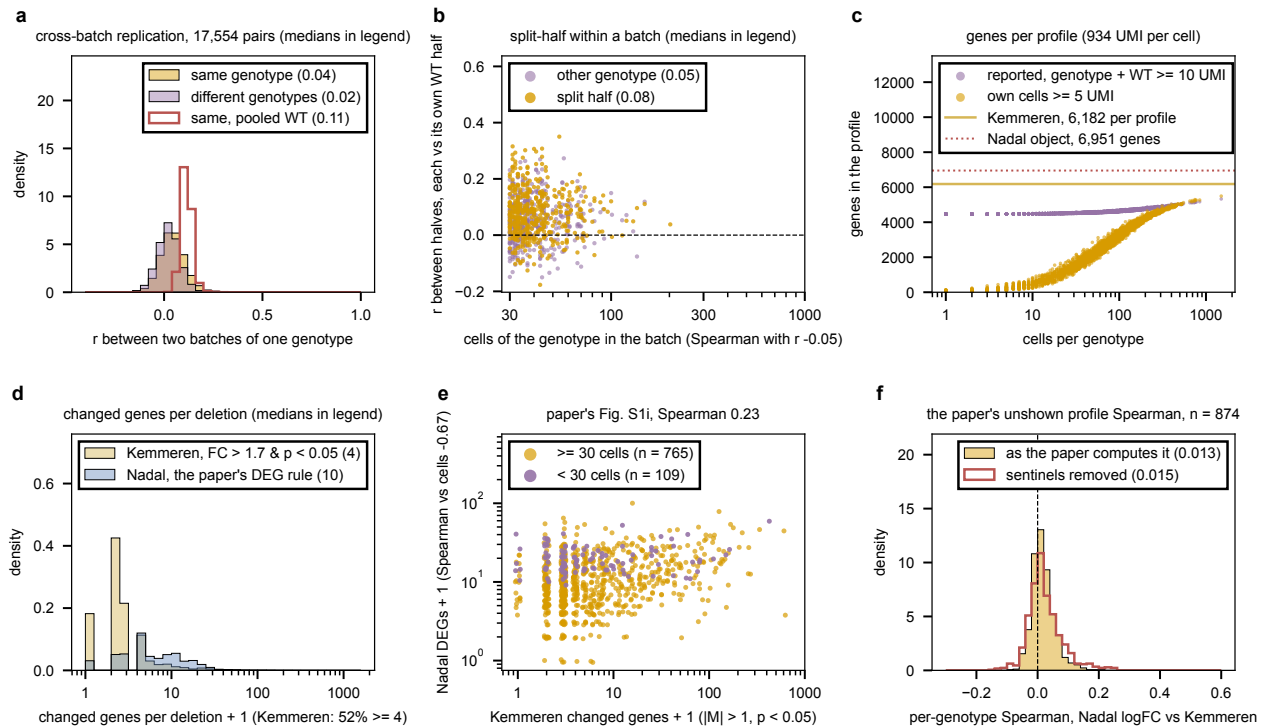


Figure 4: The same genotype measured twice agrees no better than two different genotypes, and the published agreement statistic is a cell-count artifact. (a) A genotype's profile in one capture cartridge against its profile in another, each referenced to wild-type cells from its own cartridge, with the different-genotype null beside it. The pooled-reference version of the same comparison is shown to its right, where a common batch effect lifts both. (b) The same within one cartridge, splitting the genotype's cells and the wild-type cells in half so the two profiles share no cell. (c, d) Coverage and depth: a median cell carries 934 unique molecules over 492 genes, and a median genotype has 92 cells. (e) The comparison the paper publishes, a count of changed genes against the microarray's count, which correlates at 0.23. (f) The profile correlation the paper's own released script computes and does not show, median 0.013. Its changed-gene count falls with cell number at a rank correlation of -0.67 , because a genotype with few cells has more genes at a zero mean and each one becomes a sentinel that passes the fold-change threshold. Generated by `experiments/028-knockout-expression/scripts/nadal_replication_coverage.py`.

The positive control fails too. A deletion removes its own transcript, so in a clean profile the deleted gene sits at the bottom. In the microarray data it is the single most reduced gene in **59.5%** of 1,479 strains and in the bottom one percent in 93%. In the single-cell panel it is the most reduced gene in under 1% of genotypes, and its median rank sits near the middle of the profile. Direct measurement of label purity puts roughly **40%** of the cells under a label as the genotype that label names.

Everything else in our database agrees with everything else. Five independent yeast panels, two knockout microarray studies, a knockout proteome, a natural-isolate transcriptome survey and this screen, can be compared without sharing a single strain, by asking whether the same pairs of genes move together within each panel. The four established panels agree with one another at 0.30 to 0.68. This one agrees with all of them at 0.03 to 0.10, and its gene-pair structure is nearly flat.

The lesson is three cheap checks, not an argument against single-cell screening. What went wrong is measurable from the data itself, so build the measurement in. Replicate at least one genotype across two independent barcode wells or cartridges and reference each to wild-type cells from the same one. Read the targeted gene's own transcript as a positive control, since CRISPR interference should reduce it in the cells carrying that guide. And report cells per label and label purity beside every result. The screen above would have failed all three before any biology was claimed from it.

4 The Two Decisions That Were Open ✗

4.1 Growth Media, or the Readout First ✗

Both, and the split falls along the two hosts rather than along a preference.

Growth media are the cheap axis. Cell demand grows in proportion to the number of conditions, where covering named pairs of genes grows with the square of the panel, and any route whose first barcode is a plate of wells labels conditions for nothing (fig. 1b). A model that proposes strain designs also needs both halves of its input to vary, and a dataset that varies only genotype in one medium constrains only the genotype half.

What media cannot do is multiply an assay that does not exist yet. Running four conditions through a readout whose guide detection is unmeasured produces four copies of the same unanswered question, at four times the price. So the ordering is not a judgment about which is more interesting.

- *S. cerevisiae* spends its first run on the readout, in one condition, because the library exists and q is the gate (section 2). Media come second, and by then the plate route is the way to buy them.
- *P. kudriavzevii* spends its first run on media, because it has no library to gate on. A single unperturbed condition under-uses a channel that has already been paid for, and the honest way to thicken that run is more conditions rather than genetics that do not exist yet.

On the plate route eight media ride in one protocol run for \$4,990 plus the one-time plate purchase, against \$12,550 for four media as four droplet channels (table 1). That comparison is the argument for the plate, and it only applies once the plate chemistry is known to work on this cell wall, which nobody has shown. Running the *P. kudriavzevii* first experiment on droplet across a few media, rather than on the plate across many, buys the wall answer on a platform the core supports and leaves the plate for run two.

4.2 Plasmid Copy Number ✗

Not early, with one exception worth building into the first run.

Copy number is not guide diversity, and conflating them is the trap. A cell that takes up one plasmid molecule and amplifies it to fifty copies carries fifty copies of *one* guide. Diversity comes only from the number of independent uptake events at transformation. Raising average copy number by weakening the selection marker is well established and finely tunable in yeast, over “5–100 copies per cell” [8], and it does not by itself put two guides in a cell.

Three reasons to keep the pilot low-copy. A single-guide screen is interpretable precisely because one cell means one guide, and the field builds its libraries in “typically, low-copy number vectors to prevent multiple guide plasmids in a single cell” [9]. Multiple plasmids per cell are also transient: multi-vector transformants segregate over roughly a day of outgrowth [10], and 2μ plasmids are retained in only about 60% of cells even in a laboratory strain [11], so a high-copy pilot measures a population that is changing while it is being measured. And a cell carrying two guides at unknown copy number confounds the one number the first run exists to produce, since a guide missed in such a cell and a guide missed in a single-guide cell are not the same failure.

The exception, and it is a hypothesis. Guide detection depends on how much guide transcript is present, and copy number raises that. **Hypothesis (untested):** a higher-copy vector raises q by putting more guide RNA in each cell. Nothing measures this, and it is directly testable in the run already planned, as a second channel carrying the same library on a higher-copy backbone (table 1, row two, \$8,576 for the pair rather than \$4,493). Worth doing if the budget allows, and worth skipping if the first channel returns a high q on its own.

When copy number does become the right lever. Once q is known and single-guide screening works, putting several guides in one cell is what makes combinations affordable, and the multi-plasmid route is the only way past the two-to-four guide ceiling that cloning arrays into one vector imposes. The missing measurement then is the per-cell

distribution of distinct guides under tuned delivery, which the literature does not report for yeast. That is the second pilot, not the first.

Keep the first library low-copy, and treat high copy as a detection experiment rather than a multiplexing one. Copy number buys guide transcript per cell, not guides per cell, so it cannot deliver combinations and it can confound the measurement of q . The one version worth running now is a second channel of the same library on a higher-copy backbone, purely to see whether q rises with it.

5 Running It ✗

5.1 What Each Run Must Return ✗

A run is finished when it has produced these, and not when it has produced data.

***S. cerevisiae*, one 5' channel on the existing CRISPRi library.**

1. q , the fraction of quality-controlled cells carrying exactly one identified guide. This is the number the run exists for.
2. The targeted gene's own transcript in the cells carrying its guide, against cells carrying other guides. Interference should reduce it, and this is the positive control whose absence went unnoticed in section 3.
3. Messenger RNA molecules and genes per cell, and the fraction of reads that are ribosomal.
4. Guide representation: how many of the library's guides appear at all, and how unevenly.

***P. kudriavzevii*, one channel, several media, no library.**

1. Whether the wall yields, read as cells recovered against cells loaded.
2. Messenger RNA molecules and genes per cell.
3. Ribosomal fraction, which is the largest single lever on the cost of every later experiment and is a proportion, so it is answerable at any sequencing depth.
4. Whether the media separate, which is both a biological result and the evidence that condition labels survive the protocol.

5.2 Three Things to Get Right at the Bench ✗

Count the cells on an instrument that can see yeast. *S. cerevisiae* runs about 3.5 to 5 μm across ▲ [external: vendor sizing notes; no primary in the mirror], at or below the stated floor of most brightfield counters, and 10x's own guidance is that cells under 10 μm "may be difficult to distinguish from debris" and that a fluorescent viability stain rather than trypan blue is "strongly recommended" ▲ [external: 10x CG00053]. Two instruments are validated on this organism by their makers, the Logos LUNA-FX7 and the ChemoMetec NC-203 ▲ [external: Logos and ChemoMetec application notes]. The Bio-Rad TC20 specifies a 6 μm floor ▲ [external: Bio-Rad TC20 brochure], which excludes yeast. The count sets the droplet loading rate and therefore the doublet rate, and the core named it as the single most critical step.

Replicate one genotype across two independent wells. Not a second channel and not a technical resequencing. Two wells, or two cartridges, with wild-type cells in each, so that a profile can be compared against itself with a reference from its own batch. This is the check that would have caught the published screen, it costs a handful of wells, and without it a null result later cannot be told from a broken assay.

Sequence twice, shallow then deep. A MiSeq run confirms the chemistry worked, and the questions it answers are proportions, so they need almost no depth. A NovaSeq X 10B split lane then buys 600 million read pairs for \$1,160, which is less than the MiSeq run costs and 24 times the reads. There is no earlier checkpoint than sequencing something: complementary DNA traces and library concentration do not predict whether barcoding worked [5].

5.3 What Happens Next, Under Each Outcome ✗

If q is high in *S. cerevisiae*. The library needs no rebuilding. Run a focused single-guide pilot on a metabolic sub-library at 100 cells per gene, and add growth media to it through the plate route, where they are close to free.

If q is near zero. The 5' scaffold primer does not engage a polymerase III transcript. Build the polyadenylated guide barcode and validate it on about ten arrayed strains with known knockdown phenotypes against bulk RNA sequencing of the same strains. Ground truth is only available while the design is arrayed, since a pool cannot say what was in a given cell.

If the *P. kudriavzevii* wall does not yield. The failure is in sample preparation rather than in the platform, and the next attempt varies the digestion rather than the sequencing. Nothing else in the plan depends on it, which is why it is worth running now.

If both work. The two tracks converge on one question, which is whether to spend the next budget on more perturbations per cell or on more growth media. The arithmetic favors media, and the model that would consume the data needs both, so the first crossed design is a small metabolic panel across a handful of conditions rather than a genome-scale screen in one.

Two runs, two people, about \$8,300, and every later decision depends on what they return. The *S. cerevisiae* channel decides whether a construction project stands between us and a screen. The *P. kudriavzevii* channel decides whether the assay works on that organism at all. Build the three quality checks of section 3 into both, because the published screen they come from is the design this plan would otherwise have copied.

References

- [1] Lian, J., Schultz, C., Cao, M., Hamedirad, M. & Zhao, H. Multi-functional genome-wide CRISPR system for high throughput genotype–phenotype mapping. *Nature Communications* **10**, 5794 (2019).
- [2] Brandner, J. R. *et al.* Pooled single-cell CRISPRa/i screens for functional genomics in bacteria at scale (2025).
- [3] Nadal-Ribelles, M. *et al.* A single-cell resolved genotype-phenotype map using genome-wide genetic and environmental perturbations. *Nature Communications* **16**, 2645 (2025).
- [4] Jariani, A. *et al.* A new protocol for single-cell RNA-seq reveals stochastic gene expression during lag phase in budding yeast. *eLife* **9**, e55320 (2020).
- [5] Brettner, L., Eder, R., Schmidlin, K. & Geiler-Samerotte, K. An ultra high-throughput, massively multiplexable, single-cell RNA-seq platform in yeasts. *Yeast* **41**, 242–255 (2024).
- [6] Kemmeren, P. *et al.* Large-Scale Genetic Perturbations Reveal Regulatory Networks and an Abundance of Gene-Specific Repressors. *Cell* **157**, 740–752 (2014).
- [7] Sameith, K. *et al.* A high-resolution gene expression atlas of epistasis between gene-specific transcription factors exposes potential mechanisms for genetic interactions. *BMC Biology* **13**, 112 (2015).
- [8] Lian, J., Jin, R. & Zhao, H. Construction of plasmids with tunable copy numbers in *saccharomyces cerevisiae* and their applications in pathway optimization and multiplex genome integration. *Biotechnology and Bioengineering* **113**, 24622473 (2016). <http://dx.doi.org/10.1002/bit.26004>.
- [9] Robertson, N. R., Lee, S., Tafrishi, A. & Wheeldon, I. Advances in crispr-enabled genome-wide screens in yeast. *FEMS Yeast Research* **25** (2025). <http://dx.doi.org/10.1093/femsyr/foaf013>.
- [10] Scanlon, T. C., Gray, E. C. & Griswold, K. E. Quantifying and resolving multiple vector transformants in *s. cerevisiae* plasmid libraries. *BMC Biotechnology* **9** (2009). <http://dx.doi.org/10.1186/1472-6750-9-95>.
- [11] Hays, M., Young, J. M., Levan, P. F. & Malik, H. S. A natural variant of the essential host gene *mms21* restricts the parasitic 2-micron plasmid in *saccharomyces cerevisiae*. *eLife* **9** (2020). <http://dx.doi.org/10.7554/eLife.62337>.